Notes on "An Introduction to Manifolds" by Tu

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1 Smooth Functions on Euclidean Space

In an introductory calculus course, we had worked with $\mathbb{R}^n$. $\mathbb{R}^n$ is a manifold, but it also is equipped with a natural coordinate system, unlike many other manifolds. It turns out that a lot of stuff from calculus that was done was only possible because there existed a corresponding global coordinate system. For example, it is impossible to integrate functions on a manifold in general, since integration requires a coordinate system. The objects that can be integrated are differential forms.

The goal is to recast calculus in $\mathbb{R}^n$ in a way independent of coordinates to generalize to manifolds.

1.1 Smooth vs Analytic Functions

Write coordinates on $\mathbb{R}^n$ as $x^1, \cdots, x^n$ and let $p = (p^1, \cdots, p^n)$ be a point in an open set of $U \subset \mathbb{R}^n$. Index using indices as superscripts (there is reason for this).

Definition

Let $k \in \mathbb{Z}_0^+$. A real-valued function $f : U \to \mathbb{R}^n$ is said to be C^k at $p \in U$ if its partial derivatives

$$\frac{\partial^j f}{\partial x^{i_1} \cdots \partial x^{i_j}}$$

of all orders $j \leq k$ exists and are continuous at p. The function $f : U \to \mathbb{R}^n$ is C^∞ at $p \in U$ if it is C^k for all $k \geq 0$. A vector-valued function $f : U \to \mathbb{R}^m$ is said to be C^k if all of its component functions $f^1, \cdots, f^m$ are C^k at p. We say that $f : U \to \mathbb{R}^m$ is C^k on U if it is C^k for every point $p \in U$. The definition for C^∞ for vector-valued functions are defined in a similar manner. C^∞ and smooth are equivalent.

A neighborhood of a point in $\mathbb{R}^n$ is an open set containing the point. The function f is real analytic at p if in some neighborhood of p it is equal to its

Taylor series at p, i.e.

$$f(x) = f(p) + \sum_i \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \frac{1}{2!}\sum_{i,j} \frac{\partial^2 f}{\partial x^i \partial x^j}(p)(x^i - p^i)(x^j - p^j)$$
$$+ \cdots + \frac{1}{k!}\sum_{i_1,\cdots,i_k} \frac{\partial^k f}{\partial x^{i_1}\cdots\partial x^{i_j}}(p)(x^{i_1} - p^{i_1})\cdots(x^{i_k} - p^{i_k}) + \cdots$$

All real-analytic functions are C^∞, but not all C^∞ functions are real-analytic. Consider the function

$$f(x) = \begin{cases} e^{-1/x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

It is relatively easy to show via induction that $f(x)$ is C^∞. However, note that at $x = 0$, $f(0) = 0$ and $f^{(k)}(0) = 0$ and so its Taylor series expansion should be 0. However, for $x > 0$, $f(x) = e^{-1/x} \neq 0$ and so there does not exist a neighborhood around $x = 0$ such that $f(x)$ is identical to its Taylor series.

1.2 Taylor's Theorem with Remainders

While a C^∞ is not necessarily equal to its Taylor series, there is a Taylor's theorem with remainder that is good enough. In the theorem below, the Taylor series will only consist of the constant term $f(p)$

Taylor's theorem with remainders

Let f be C^∞ function on an open convex subset $U \subset \mathbb{R}^n$. Let $p = (p^1, \cdots, p^n) \in U$. Then there exists functions $g_1(x), \cdots, g_n(x) \in C^\infty(U)$ such that

$$f(x) = f(p) + \sum_{i=1}^n (x^i - p^i) g_i(x)$$

where

$$g_i(p) = \frac{\partial f}{\partial x^i}(p)$$

Proof: For any point $x \in U$, there exists a line segment entirely in U which connects x to p, given by $p + t(x - p)$, for $0 \leq t \leq 1$. Thus, $f(p + t(x - p))$ exists and is C^∞ for all points along the line segment. Taking the derivative of this function with respect to t, we get

$$\frac{d}{dt} f(p + t(x - p)) = \sum_{i=1}^n (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x - p))$$

Integrating with respect to t from 0 to 1, we get

$$\int_0^1 \frac{d}{dt} f(p + t(x - p)) dt = \int_0^1 \sum_{i=1}^n (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x - p)) dt$$

The LHS evaluates to

$$\int_0^1 \frac{d}{dt} f(p + t(x-p))dt = f(p + t(x-p))\Big|_0^1$$

and the RHS evaluates to

$$\int_0^1 \sum_{i=1}^n (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x-p))dt = \sum_{i=1}^n (x^i - p^i) \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt$$

If we define $g_i(x)$ to be

$$g_i(x) = \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt$$

we see that $g_i(x)$ is C^∞ as f is also C^∞. Combining the results, we have that

$$f(x) - f(p) = \sum_{i=1}^n (x^i - p^i) g_i(x)$$

Now finally, notice that when $x = p$, $p + t(x-p) = p + t(p-p) = p$ and so

$$g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p) \int_0^1 dt = \frac{\partial f}{\partial x^i}(p)$$

Exercise 1.2

Let $f(x)$ be

$$f(x) = \begin{cases} e^{-1/x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

Show by induction that for $x > 0$ and $k \geq 0$, the kth derivative $f^{(k)}(x)$ is of the form $p_{2k}(1/x)e^{-1/x}$ for some polynomial $p_{2k}(y)$ of degree $2k$ in y. Prove that f is C^∞ on $\mathbb{R}$ and that $f^{(k)}(0) = 0$ for all $k \geq 0$

Clearly, for $x > 0$, $f(x) = e^{-1/x}$. In the base case $n = 0$, $f(x) = e^{-1/x}$ and so $p_0 = 1$ is a polynomial of degree 0. Now suppose that for some $n = k$, $f^{(k)}(x) = p_{2k}(1/x)e^{-1/x}$ for some polynomial $p_{2k}(y)$ of degree $2k$ in y. Then

$$\begin{aligned} f^{(k+1)}(x) &= \left(p'_{2k}(1/x) \cdot -\frac{1}{x^2} e^{-1/x} \right) + \left(p_{2k}(1/x) e^{-1/x} \cdot \frac{1}{x^2} \right) \\ &= e^{-1/x} \left(1/x\right)^2 \left(p_{2k}(1/x) - p'_{2k}(1/x)\right) \end{aligned}$$

Letting $y = 1/x$, we see that

$$\begin{aligned} e^{-1/x} \left(1/x\right)^2 \left(p_{2k}(1/x) - p'_{2k}(1/x)\right) &= e^{-1/x} y^2 \left(p_{2k}(y) - p'_{2k}(y)\right) \\ &= e^{-1/x} p_{2k+2}(y) = e^{-1/x} p_{2(k+1)}(1/x) \end{aligned}$$

Self-evidently, when $x < 0$, $f(x) = 0$ is C^∞ and likewise, when $x > 0$, $f(x) = e^{-1/x}$ is C^∞. The only contention arises when dealing with $x = 0$, as one must show all the derivatives of $f(x)$ is continuous there.

To show that $f(x)$ is C^∞ at 0, we need to show that for $k \geq 0$

$$\lim_{x \to 0} f^{(k)}(x) = 0$$

Clearly, since $f(x) = 0$ for $x < 0$, we have that

$$\lim_{x \to 0^-} f^{(k)}(x) = 0$$

Since $f(x) = e^{-1/x}$, $f^{(k)}(x) = p_{2k}(1/x)e^{-1/x}$ for some polynomial $p_{2k}(y)$ of degree y.

$$\lim_{x \to 0^+} f^{(k)}(x) = \lim_{y \to \infty} p_{2k}(y)e^{-y} = 0$$

Therefore, $f(x)$ is C^∞ at 0.

Exercise 1.6

Prove that if $f : \mathbb{R}^2 \to \mathbb{R}$ is C^∞, then there exists functions g_{11}, g_{12}, g_{22} such that

$$\begin{aligned} f(x,y) = f(0,0) + \frac{\partial f}{\partial x}(0,0)x + \frac{\partial f}{\partial y}(0,0)y + x^2 g_{11}(x,y) \\ + xy g_{12}(x,y) + y^2 g_{22}(x,y) \end{aligned}$$

Since we know that f is C^∞, we know from Taylor's theorem with Remainder that there exists C^∞ functions g_1, g_2 such that

$$\begin{aligned} f(x,y) &= f(0,0) + (x-0)g_1(x,y) + (y-0)g_2(x,y) \\ &= f(0,0) + x g_1(x,y) + y g_2(x,y) \end{aligned}$$

and

$$\begin{aligned} g_1(0,0) &= \frac{\partial f}{\partial x}(0,0) \\ g_2(0,0) &= \frac{\partial f}{\partial y}(0,0) \end{aligned}$$

Applying Taylor's theorem with remainder to the function $g_1(x,y)$, we see that there exists functions h_{11}, h_{12} in C^∞ such that

$$\begin{aligned} g_1(x,y) &= g_1(0,0) + (x-0)h_{11}(x,y) + (y-0)h_{12}(x,y) \\ &= \frac{\partial f}{\partial x}(0,0) + x h_{11}(x,y) + y h_{12}(x,y) \end{aligned}$$

Similarly, applying it to $g_2(x, y)$, we see that there exists functions h_{21}, h_{22} in C^∞ such that

$$\begin{aligned} g_2(x, y) &= g_2(0, 0) + (x - 0)h_{21}(x, y) + (y - 0)h_{22}(x, y) \\ &= \frac{\partial f}{\partial y}(0, 0) + xh_{21}(x, y) + yh_{22}(x, y) \end{aligned}$$

Plugging these results in, we have that

$$\begin{aligned} f(x, y) &= f(0, 0) + x\left(\frac{\partial f}{\partial x}(0, 0) + xh_{11}(x, y) + yh_{12}(x, y)\right) \\ &\quad + y\left(\frac{\partial f}{\partial y}(0, 0) + xh_{21}(x, y) + yh_{22}(x, y)\right) \\ f(x, y) &= f(0, 0) + \frac{\partial f}{\partial x}(0, 0)x + \frac{\partial f}{\partial y}(0, 0)y + x^2h_{11}(x, y) \\ &\quad + xy\left(h_{12}(x, y) + h_{21}(x, y)\right) + y^2h_{22}(x, y) \end{aligned}$$

Now let $g_{11}(x, y) = h_{11}(x, y)$, $g_{12}(x, y) = h_{12}(x, y) + h_{21}(x, y)$, and $g_{22}(x, y) = h_{22}(x, y)$. This completes the proof.

Exercise 1.7

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a C^∞ function with

$$f(0, 0) = \frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$$

Define a function $g : \mathbb{R}^2 \to \mathbb{R}$ by

$$g(t, u) = \begin{cases} \dfrac{f(t, tu)}{t}, & t \neq 0 \\ 0, & t = 0 \end{cases}$$

Prove that $g(t, u)$ is C^∞ for $(t, u) \in \mathbb{R}^2$

Recall from the earlier exercise 1.6 that since f is C^∞, then there exists functions g_{11}, g_{12}, g_{22} such that

$$\begin{aligned} f(x, y) &= f(0, 0) + \frac{\partial f}{\partial x}(0, 0)x + \frac{\partial f}{\partial y}(0, 0)y + x^2g_{11}(x, y) \\ &\quad + xyg_{12}(x, y) + y^2g_{22}(x, y) \end{aligned}$$

Plugging in the earlier conditions, we have that

$$f(x, y) = x^2g_{11}(x, y) + xyg_{12}(x, y) + y^2g_{22}(x, y)$$

For $x = t, y = tu$, we have that

$$f(t, tu) = t^2g_{11}(t, tu) + t^2ug_{12}(t, tu) + (tu)^2g_{22}(t, tu)$$

and so finally, we have that

$$\begin{aligned}\frac{f(t,tu)}{t} &= tg_{11}(t,tu) + tug_{12}(t,tu) + tu^2 g_{22}(t,tu)\\ &= t\left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right)\end{aligned}$$

Recall the earlier definition of $g(t,u)$. The derivative of this function with respect to t is

$$\begin{aligned}\frac{\partial g}{\partial t}(t,u) &= \left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right)\\ +t\left(\frac{\partial g_{11}}{\partial x}(t,tu)\frac{\partial x}{\partial t} + \frac{\partial g_{11}}{\partial y}(t,tu)\frac{\partial y}{\partial t}\right. &+ u\left.\left(\frac{\partial g_{12}}{\partial x}(t,tu)\frac{\partial x}{\partial t} + \frac{\partial g_{12}}{\partial y}(t,tu)\frac{\partial y}{\partial t}\right)\right)\\ &+tu^2\left(\frac{\partial g_{22}}{\partial x}(t,tu)\frac{\partial x}{\partial t} + \frac{\partial g_{22}}{\partial y}(t,tu)\frac{\partial y}{\partial t}\right)\\ &= \left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right)\\ +t\left(\frac{\partial g_{11}}{\partial x}(t,tu) + u\frac{\partial g_{11}}{\partial y}(t,tu)\right. &+ u\left.\left(\frac{\partial g_{12}}{\partial x}(t,tu) + u\frac{\partial g_{12}}{\partial y}(t,tu)\right)\right)\\ &+tu^2\left(\frac{\partial g_{22}}{\partial x}(t,tu) + u\frac{\partial g_{22}}{\partial y}(t,tu)\right)\end{aligned}$$

Similarly, the derivative of g with respect to u is

$$\begin{aligned}\frac{\partial g}{\partial u}(t,u) &= t\left(\frac{\partial g_{11}}{\partial x}(t,tu)\frac{\partial x}{\partial u} + \frac{\partial g_{11}}{\partial y}(t,tu)\frac{\partial y}{\partial u} + g_{12}(t,tu) + u\left(\frac{\partial g_{12}}{\partial x}(t,tu)\frac{\partial x}{\partial u} + \frac{\partial g_{12}}{\partial y}(t,tu)\frac{\partial y}{\partial u}\right)\right)\\ &+t\left(2ug_{22}(t,tu) + u^2\left(\frac{\partial g_{22}}{\partial x}(t,tu)\frac{\partial x}{\partial u} + \frac{\partial g_{22}}{\partial y}(t,tu)\frac{\partial y}{\partial u}\right)\right)\\ &= t\left(t\frac{\partial g_{11}}{\partial y}(t,tu) + g_{12}(t,tu) + u\left(t\frac{\partial g_{12}}{\partial y}(t,tu)\right) + 2ug_{22}(t,tu) + u^2\left(t\frac{\partial g_{22}}{\partial y}(t,tu)\right)\right)\end{aligned}$$

Notice where $t \neq 0$

$$g(t,u) = \frac{f(t,tu)}{t} = t\left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right)$$

is C^∞ for all points (t,u) where $t \neq 0$ since all the component parts are also C^∞. It is also continuous at $t = 0$ since

$$\lim_{t\to 0} t\left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right) = 0$$

since g_{11}, g_{12}, g_{22} are C^∞ and so must converge to a definite value. Finally, notice that the partial derivatives

$$\nabla g(t,u) = \left(\frac{\partial g}{\partial t}(t,u), \frac{\partial g}{\partial u}(t,u)\right)$$

are composed of functions that at the limit is C^∞ at $t = 0$ and so the partial derivatives are C^∞ at $t = 0$.

Tangent vectors in $\mathbb{R}^n$ as Derivations

We typically express a vector v at a point $p \in \mathbb{R}^n$ as a column of numbers

$$v = \begin{bmatrix} v^1 \\ \vdots \\ v^n \end{bmatrix}$$

or as a row of numbers bounded by angular brackets

$$v = \langle v^1, \cdots, v^n \rangle$$

Recall that a secant plane to a surface in $\mathbb{R}^3$ is given by three non-collinear points on a surface, and as the three points approach a point p on the surface, the plane at the limit position of the secant plane is called the tangent plane at p. Intuitively, we can understand this as the plane simply "touching" the surface at this point. A vector at p is tangent to the surface if it lies in the tangent plane at p.

Directional Derivative

In $\mathbb{R}^n$, we can visualize the tangent space $T_p(\mathbb{R}^n)$ at p as the vector space of all vectors emanating from p. Since we can express the vector space with a list of n numbers, it is identical to $\mathbb{R}^n$. To distinguish between points and vectors then, we let a point $p \in \mathbb{R}^n$ be written as $p = \left(p^1, \cdots, p^n\right)$ and a vector $v \in T_p(\mathbb{R}^n)$ be written as

$$v = \begin{bmatrix} v^1 \\ \vdots \\ v^n \end{bmatrix} \quad \text{or} \quad v = \langle v^1, \cdots, v^n \rangle$$

Typically, the standard basis for $\mathbb{R}^n$ or $T_p(\mathbb{R}^n)$ is denoted by $e_1, \cdots, e_n$. Then, we can express $v \in T_p(\mathbb{R}^n)$ as $v = \sum v^i e_i$ for some $v^i \in \mathbb{R}$. Elements of $T_p(\mathbb{R}^n)$ are called tangent vectors at p in $\mathbb{R}^n$.

The line through a point $p = \left(p^1, \cdots, p^n\right)$ with a direction $v = \langle v^1, \cdots, v^n \rangle$ has parametrization

$$c(t) = \left(p^1 + tv^1, \cdots, p^n + tv^n\right)$$

If f is C^∞ at p and v is a tangent vector at p, then the directional derivative of f in the direction of v at p is defined to be

$$D_v f = \lim_{t \to 0} \frac{f(c(t)) - f(p)}{t} = \left.\frac{d}{dt}\right|_{t=0} f(c(t))$$

By the chain rule,

$$D_v f = \sum_{i=1}^{n} \frac{dc^i}{dt}(0) \frac{\partial f}{\partial x^i}(p)$$

We know that $c'(t) = v$ and so

$$D_v f = \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}(p) = v \cdot \nabla f(p)$$

The reason the notation $D_v f$ is used, instead of $D_v f(p)$, is that the point p is already encoded in the vector v, since $v \in T_p\mathbb{R}^n$. v is anchored at p. Thus, the directional derivative is inherently taken at p. Moreover, $D_v f$ is understood as a number, the directional derivative at p, and not a function of p.

We write

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p$$

to be the map that sends a function f to the number $D_v f$. The association $v \mapsto D_v$ of the directional derivative D_v to the tangent vector v gives a way to characterize tangent vectors as operators on functions.

Germs of Functions

Equivalence Relation

A relation on a set S is a subset $R \subset S \times S$. R is an equivalence relation if it satisfies for all $x, y, z \in S$

1. (Reflexivity) $x \sim x$
2. (Symmetry) $x \sim y$ if and only if $y \sim x$
3. (Transitivity) $x \sim y$ and $y \sim z$ implies $x \sim z$

If two functions agree on some neighborhood of a point p, they would have the same directional derivatives at p, suggesting that an equivalence relation can be defined on the C^∞ functions defined on some neighborhood of p.

Germs of Functions

Consider the set of all pairs (f, U), where U is a neighborhood of p and $f : U \to \mathbb{R}$ is C^∞. We say that two pairs (f, U) and (g, V) are equivalent to each other if there exists an open set $W \subset U \cap V$ containing p such that $f = g$ when restricted to W. The equivalence class of (f, U) is called the germ of f at p. We write $C_p^\infty(\mathbb{R}^n)$ (or C_p^∞ if there is no possibility of confusion) for the set of all germs of C^∞ functions on $\mathbb{R}^n$ at p.

Consider the function

$$f(x) = \frac{1}{1-x}$$

over the domain $\mathbb{R} - \{1\}$ and

$$g(x) = 1 + x + x^2 + x^3 + \cdots$$

with the domain $(-1, 1)$. f and g have the same germ at any point $p \in (-1, 1)$.

Algebra

An algebra over a field K is a vector space A over K with the multiplication map

$$\mu : A \times A \to A$$

denoted typically by $\mu(a, b) = a \cdot b$, such that for all $a, b, c \in A$ and $r \in K$

1. (Associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. (Distribution) $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + a \cdot b$
3. (Homegeneuity) $r(a \cdot b) = (ra) \cdot b = a \cdot (rb)$

This is equivalent to stating that an algebra over a field K is a ring A that is also a vector space over K such that the ring multiplication satisfies the homogeneity condition.

Linear Operator

A map $L : V \to W$ between vector spaces over a field K is called a linear operator if for any $r \in K$ and $u, v \in V$

1. $L(u + v) = L(u) + L(v)$
2. $L(rv) = rL(v)$

If A and A' are algebras over a field K, then an algebra homomorphism is a linear map $L : A \to A'$ that preserves the algebra multiplication, i.e. $L(a \cdot b) = L(a) \cdot L(b)$ for all $a, b \in A$.

The addition and multiplication of functions induces the corresponding operations on C_p^∞, making it into an algebra over $\mathbb{R}$.

Derivation at a Point

For each tangent vector v at a point $p \in \mathbb{R}^n$, the directional derivative at p gives a map of real vector spaces

$$D_v : C_p^\infty \to \mathbb{R}$$

Consider two functions $f, g \in C_p^\infty$ and a scalar $r \in \mathbb{R}$. Recall that since

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p$$

we have that

$$\begin{aligned} D_v(f+g) = \sum_{i=1}^{n} v^i \frac{\partial(f+g)}{\partial x^i}(p) &= \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}(p) + \sum_{i=1}^{n} v^i \frac{\partial g}{\partial x^i}(p) \\ &= D_v f + D_v g \end{aligned}$$

Similarly,

$$D_v(rf) = \sum_{i=1}^n v^i \frac{\partial(rf)}{\partial x^i}(p) = r\sum_{i=1}^n v^i \frac{\partial f}{\partial x^i}(p) = rD_v f$$

Therefore, D_v is a linear map (D_v is $\mathbb{R}$-linear). Furthermore, we know that D_v satisfies the Leibniz rule

$$D_v(fg) = (D_v f)g(p) + f(p)D_v g$$

since

$$\begin{aligned}D_v(fg) &= \sum_{i=1}^n v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left(\frac{\partial f}{\partial x^i}(p)g(p) + f(p)\frac{\partial g}{\partial x^i}(p)\right)\\ &= g(p)\sum_{i=1}^n v^i \frac{\partial f}{\partial x^i}(p) + f(p)\sum_{i=1}^n v^i \frac{\partial g}{\partial x^i}(p) = (D_v f)g(p) + f(p)(D_v g)\end{aligned}$$

Any linear map $D : C_p^\infty \to \mathbb{R}$ that satisfies the Leibniz rule is referred to as a *derivation at* p or a *point-derivation* of C_p^∞. Denote the set of all derivations at p by $\mathcal{D}_p(\mathbb{R}^n)$. This space is a vector space.

We know that all directional derivatives at p are all derivations at p, so there exists a map

$$\phi : T_p(\mathbb{R}^n) \to \mathcal{D}_p(\mathbb{R}^n)$$

given by

$$v \mapsto D_v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\bigg|_p$$

We can deduce that ϕ is a linear map since for vectors $v, w \in T_p(\mathbb{R}^n)$ and $r \in \mathbb{R}$

$$\begin{aligned}\phi(v+w) = D_{v+w} &= \sum_{i=1}^n (v+w)^i \frac{\partial}{\partial x^i}\bigg|_p = \sum_{i=1}^n (v^i + w^i)\frac{\partial}{\partial x^i}\bigg|_p\\ &= \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\bigg|_p + \sum_{i=1}^n w^i \frac{\partial}{\partial x^i}\bigg|_p = D_v + D_w = \phi(v) + \phi(w)\end{aligned}$$

and

$$\begin{aligned}\phi(rw) = D_{rw} &= \sum_{i=1}^n (rw)^i \frac{\partial}{\partial x^i}\bigg|_p = \sum_{i=1}^n r(w^i)\frac{\partial}{\partial x^i}\bigg|_p\\ &= r\sum_{i=1}^n w^i \frac{\partial}{\partial x^i}\bigg|_p = rD_w = r\phi(w)\end{aligned}$$

Lemma 2.1

If D is a point-derivation of C_p^∞, then $D(c) = 0$ for any constant function c.
Proof: Since every point-derivation is $\mathbb{R}$-linear, we know that $D(c) = cD(1)$, and so what we need to show is that $D(1) = 0$. We know by Leibniz rule that

$$D(1) = D(1 \cdot 1) = D(1) \cdot 1 + 1 \cdot D(1) = 2D(1)$$

and so $D(1) = 0$. Thus, $D(c) = 0$.

The Kronecker delta δ is a convenient notation used defined in the following manner

$$\delta_i^j = \begin{cases} 1 & i = j, \\ 0 & i \neq j \end{cases}$$

Theorem 2.2

The linear map $\phi : T_p(\mathbb{R}^n) \to \mathcal{D}_p(\mathbb{R}^n)$ given by $v \mapsto D_v$ is an isomorphism of vector spaces.
Proof: To show that this is an isomorphism, we have to show that it is bijective and that it is a linear map. We have already shown it is a linear operator, so now, we must show it is bijective.

To show that it is injective, we need to show that $\phi(a) = \phi(b)$ implies $a = b$. By using the linear properties of ϕ, we have that

$$\phi(a) - \phi(b) = \phi(a - b) = 0$$

and so $a - b \in \ker(\phi)$. To show that $a = b$, we need to show that the only vector in the kernel of ϕ is the zero vector. Suppose that for some vector $v \in T_p(\mathbb{R}^n)$, we have $D_v = 0$, i.e. v is a vector in the kernel of ϕ. Let $x^j : \mathbb{R}^n \to \mathbb{R}$ be the coordinate function sending a point q to its coordinate q^j. Applying D_v to the coordinate function, we have that

$$D_v x^j = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\bigg|_p x^j = \sum_{i=1}^n v^i \delta_i^j = v^j = 0$$

Therefore, $v = 0$ and so the kernel of ϕ consists solely of the zero vector, meaning ϕ is injective.

To show that it is surjective, let D be a derivation at p, i.e. $D \in \mathcal{D}_p(\mathbb{R}^n)$ and let (f, V) be a representative of a germ in C_p^∞. By making V as small as necessary, we can assume that V is an open ball. By Taylor's Theorem with remainder, we know there exists C^∞ functions $g_1(x), \cdots, g_n(x)$ in a neighborhood of p such that

$$f(x) = f(p) + \sum_{i=1}^n (x^i - p^i) g_i(x), \quad g_i(p) = \frac{\partial f}{\partial x^i}(p)$$

Apply D to both sides

$$\begin{aligned} Df(x) = D\left(f(p) + \sum_{i=1}^{n}(x^i - p^i)g_i(x)\right) &= Df(p) + D\left(\sum_{i=1}^{n}(x^i - p^i)g_i(x)\right) \\ &= Df(p) + \sum_{i=1}^{n} D\big((x^i - p^i)g_i(x)\big) \end{aligned}$$

From Leibniz rule, we know that

$$\begin{aligned} D\big((x^i - p^i)g_i(x)\big) &= D(x^i - p^i)g_i(p) + (p^i - p^i)Dg_i(x) \\ &= (Dx^i)g_i(p) - (Dp^i)g_i(p) \end{aligned}$$

Since $f(p)$ and p^i are constants, $Df(p) = Dp^i = 0$ and so

$$Df(x) = \sum_{i=1}^{n}(Dx^i)g_i(p) = \sum_{i=1}^{n}(Dx^i)\frac{\partial f}{\partial x^i}(p)$$

Thus, there exists a vector $v = \langle Dx^1, \cdots, Dx^n\rangle$ such that $D = D_v$. *Intuitively, a derivation D at a point p is completely determined by how it acts on the coordinate functions $x^1, \cdots, x^n$. The values Dx^i give the components of the vector v, showing that D is just the directional derivative in that direction.*

This theorem allows us to identify the tangent vectors at p with the derivations at p. Under the isomorphism $T_p(\mathbb{R}^n) \cong \mathcal{D}_p(\mathbb{R}^n)$, the standard basis $e_1, \cdots, e_n$ for $T_p(\mathbb{R}^n)$ corresponds to the set $\partial/\partial x^1|_p, \cdots, \partial/\partial x^n|_p$ of partial derivatives. Thus, we can write a tangent vector $v = \langle v^1, \cdots, v^n\rangle = \sum v^i e_i$ as

$$v = \sum v^i \frac{\partial}{\partial x^i}\bigg|_p$$

While $T_p(\mathbb{R}^n)$ is isomorphic to $\mathcal{D}_p(\mathbb{R}^n)$, there is a reason the latter is preferred over the former, which is that a vector $v \in T_p(\mathbb{R}^n)$ depends on a choice of coordinates to describe it, while a derivation $D \in \mathcal{D}_p(\mathbb{R}^n)$ is independent of coordinates and one solely relies on how it impacts functions to determine its behavior.

Vector Fields

A *vector field* X on an open subset $U \subset \mathbb{R}^n$ is a function that assigns to each point p a tangent vector X_p in $T_p(\mathbb{R}^n)$. Since $T_p(\mathbb{R}^n)$ is isomorphic to $\mathcal{D}_p(\mathbb{R}^n)$, it is convenient in this instance to use the basis $\{\partial/\partial x^i|_p\}$. Thus the vector X_p is a linear combination

$$X_p = \sum a^i(p)\frac{\partial}{\partial x^i}\bigg|_p$$

We can omit p, thus expressing X as

$$X = \sum a^i \frac{\partial}{\partial x^i}$$

where a^i are functions on U. We say the vector field X is C^∞ on U if the coefficient functions a^i are C^∞ on U.

Consider the space $\mathbb{R}^2 - \{0\}$ and let $p = (x, y)$. Then we can define a vector field X on this space by

$$X = \frac{-y}{\sqrt{x^2+y^2}} \frac{\partial}{\partial x} + \frac{x}{\sqrt{x^2+y^2}} \frac{\partial}{\partial y} = \left\langle \frac{-y}{\sqrt{x^2+y^2}}, \frac{x}{\sqrt{x^2+y^2}} \right\rangle$$

The vector X_p at a point p is a unit vector on a circle of radius $\sqrt{x^2+y^2}$ perpendicular to it going counter.

For another example, on the space $\mathbb{R}^2$, consider the vector field

$$X = \langle x, -y \rangle$$

The vector field, when visualized by plotting various vectors X_p attached to different points p, looks to move towards the x axis and away from the y axis.

One can identify the vector fields on U with column vectors of C^∞ functions on U, that is

$$X = \sum a^i \frac{\partial}{\partial x^i} \iff \begin{bmatrix} a^1 \\ \vdots \\ a^n \end{bmatrix}$$

Notice here that it is very similar to the reinterpretation of the vector

$$v = \sum v^i \frac{\partial}{\partial x^i}\bigg|_p$$

with v^i (a number corresponding to a point p) changing to a^i (a function corresponding to U, becoming a number only when the specific point is chosen).

The set of C^∞ functions equipped with the standard addition function obviously forms a ring, typically denoted by $C^\infty(U)$ or $\mathcal{F}(U)$. We will demonstrate it is a ring in Exercise 2.2.

Tu doesn't make it clear that he is no longer working with $C^\infty(U, \mathbb{R}^m)$ functions, i.e. functions from $U \subset \mathbb{R}^n$ to $\mathbb{R}^m$, and has shifted to $C^\infty(U, \mathbb{R})$, i.e. functions from $U \subset \mathbb{R}^n$ to $\mathbb{R}$. This shift is important to keep in mind to understand the context. Importantly, without keepining this in mind, the only way to establish that $C^\infty(U, \mathbb{R}^m)$ is a ring is to utilize component-wise multiplication, a very controversial choice that I personally found uncomfortable.

Multiplication of vector fields by functions on U is defined pointwise

$$(fX)_p = f(p)X_p, \quad p \in U$$

Since $X = \sum a^i \partial/\partial x^i$ is a C^∞ vector field and f is a C^∞ function on U, $fX = \sum (fa^i)\partial/\partial x^i$ is also a vector field. Thus, the set of all C^∞ vector fields on U, denoted by $\mathbb{X}(U)$, is not just a vector space over $\mathbb{R}$ but also a module over the ring $C^\infty(U)$ (but crucially, it is NOT a vector space over $C^\infty(U)$ since $C^\infty(U)$ is not a field).

Module

If R is a commutative ring (not rng), then a (left) R-module is an abelian group A with scalar multiplication

$$\mu : R \times A \to A$$

typically denoted $\mu(r, a) = ra$, such that for $r, s \in R$ and $a, b \in A$

1. (Associativity) $(rs)a = r(sa)$
2. (Identity) If 1 is the multiplicative identity in R, then $1a = a$
3. (Distributivity) $(r + s)a = ra + sa$ and $r(a + b) = ra + rb$

Module Homomorphism

Let A and A' be R-modules. An R-module homomorphism between A and A' is a map $f : A \to A'$ that preserves addition and scalar multiplication, i.e. for all $a, b \in A$ and $r \in R$

1. $f(a + b) = f(a) + f(b)$
2. $f(ra) = rf(a)$

Vector Fields as Derivations

If X is a C^∞ vector field on an open subset $U \subset \mathbb{R}^n$ and f is C^∞ function on U, then we define a new function Xf on U by

$$(Xf)(p) = X_p f, \quad p \in U$$

Recall, though, that $X = \sum a^i \partial/\partial x^i$, meaning

$$(Xf)(p) = \sum a^i(p) \frac{\partial f}{\partial x^i}(p)$$

and so

$$Xf = \sum a^i \frac{\partial f}{\partial x^i}$$

a C^∞ function on U. We can thus interpret the vector field X as an $\mathbb{R}$-linear map

$$\begin{gathered} C^\infty(U) \mapsto C^\infty(U) \\ f \mapsto Xf \end{gathered}$$

And from here, we can show that X satisfies the Leibniz rule, thus establishing that it is a derivation.

Leibniz rule for vector fields

Given that X is a C^∞ vector field and f, g are C^∞ functions on an open set $U \subset \mathbb{R}^n$, then $X(fg)$ satisfies the Leibniz rule

$$X(fg) = (Xf)g + f(Xg)$$

Proof: $X(fg)$ is defined on U by

$$(X(fg))(p) = X_p(fg)$$

for all $p \in U$. We know that

$$\begin{aligned}
X_p(fg) &= \sum a^i(p)\frac{\partial(fg)}{\partial x^i}(p) = \sum a^i(p)\left(\frac{\partial f}{\partial x^i}(p)g(p) + f(p)\frac{\partial g}{\partial x^i}(p)\right) \\
&= \sum\left(a^i(p)\frac{\partial f}{\partial x^i}(p)\right)g(p) + \sum\left(a^i(p)\frac{\partial g}{\partial x^i}(p)\right)f(p) \\
&= \left(\sum a^i(p)\frac{\partial f}{\partial x^i}(p)\right)g(p) + \left(\sum a^i(p)\frac{\partial g}{\partial x^i}(p)\right)f(p) \\
&= (X_pf)g(p) + f(p)(X_pg)
\end{aligned}$$

and so

$$X(fg) = (Xf)g + f(Xg)$$

Generalization of the Concept of Derivation

If A is an algebra over a field K, the derivation of A is a K-linear map $D : A \to A$ such that D satisfies the Leibniz rule

$$D(ab) = (Da)b + a(Db)$$

for all $a, b \in A$. The set of all derivations of A, denoted by $\mathrm{Der}(A)$, is closed under addition and scalar multiplication.

We have just shown that a C^∞ vector field X gives rise to a derivation of the algebra $C^\infty(U)$ (and not just C_p^∞). Now consider the map

$$\begin{aligned}
\phi : \mathbb{X} &\to \mathrm{Der}(C^\infty(U)) \\
X &\mapsto (f \mapsto (Xf))
\end{aligned}$$

Recall earlier that the initial conception of the vector field was a function that assigns to each point $p \in U$ a tangent vector $v \in T_p(\mathbb{R}^n)$. What this map informs us is that just like how we can identify tangent vectors v at point p with the point-derivations at C_p^∞ (as $T_p(\mathbb{R}^n)$ is isomorphic to $\mathcal{D}_p(\mathbb{R}^n)$), so to can vector fields on an open set U be identified with the derivations of the algebra $C^\infty(U)$ (we can treat it sort of like a *flow* overlapping the space, how much *work* is done by the vector field as an object traverses a path on U).

Comment: Tu doesn't explicitly contrasts Der($C^\infty(U)$) with the set of all point-wise operators formed by arbitrarily gluing together point-derivations. However, we can construct a counterexample to show where they differ. Consider the space $\mathbb{R}$. At each point $p \in \mathbb{R}$, we select a point derivation $\tilde{D}(p)$ via the following assignment

$$\tilde{D}(p) = \begin{cases} \dfrac{\partial}{\partial x}, & p \in \mathbb{Q} \\ -\dfrac{\partial}{\partial x}, & p \notin \mathbb{Q} \end{cases}$$

We are entirely free to glue together all these point-derivations to get a point-wise operator $\tilde{D} \in \prod_{p \in \mathbb{R}} \mathcal{D}_p(\mathbb{R})$. However, this operator is not an element in Der($C^\infty(U)$), because if it did, its component function $a(p)$ would have to be C^∞, but here we see that this is not the case.

Exercise 2.1

Let $X = x\partial/\partial x + y\partial/\partial y$ be a vector field on $\mathbb{R}^3$ and let $f(x,y,z) = x^2 + y^2 + z^2$ be a function on $\mathbb{R}^3$. Compute Xf

Recall that Xf is given by

$$Xf = \sum a^i \frac{\partial f}{\partial x^i}$$

We see that $a^1 = x$, $a^2 = y$, and $a^3 = 0$, and so

$$X_p f = x \frac{\partial f}{\partial x}\bigg|_p + y \frac{\partial f}{\partial y}\bigg|_p + 0 \frac{\partial f}{\partial z}\bigg|_p = x \cdot 2x + y \cdot 2y = 2\left(x^2 + y^2\right)$$

Exercise 2.2

Define addition, multiplication, and scalar multiplication in C_p^∞. Prove that addition in C_p^∞ is commutative.

Recall that C_p^∞ is the set of all germs of C^∞ functions on $\mathbb{R}^n$ at p. Furthermore, recall that two pairs (f, U) and (g, V) (where U and V are neighborhoods of p) are equivalent to each other if there exists an open set $W \subset U \cap V$ containing p such that $f|_W = g|_W$, and that the equivalence class of such pairs is called the germ of f at p.

Consider two germs $[f], [g] \in C_p^\infty$. We know that there exists pairs

$$\begin{aligned} (f_1, U_1) &\in [f] \\ (g_1, V_1) &\in [g] \end{aligned}$$

and since U_1 and V_1 are neighborhoods of p, we know there exists a nonempty set $W_1 \subset U_1 \cap V_1$ containing p. We define $[f] + [g]$ to be the equivalence class of all pairs equivalent to $(f_1|_{W_1} + g_1|_{W_1}, W_1)$. In a similar manner, we define $[f] \cdot [g]$ to be the equivalence class of all pairs equivalent to $(f_1|_{W_1} \cdot g_1|_{W_1}, W_1)$

To show that addition and multiplication is well-defined, consider two other pairs

$$(f_2, U_2) \in [f]$$
$$(g_2, V_2) \in [g]$$

Since U_2 and V_2 are neighborhoods of p, we know there exists a nonempty set $W_2 \subset U_2 \cap V_2$ containing p. Finally, since both W_1 and W_2 are neighborhoods of p, there must exist a nonempty set $W \subset W_1 \cap W_2$ containing p. Since both (f_1, U_1) and (f_2, U_2) are elements of the same germ of f at p, we know that $f_1|_U = f_2|_U$ for some set U. Similarly, since both (g_1, V_1) and (g_2, V_2) are elements of $[g]$, we know that $g_1|_V = g_2|_V$ for some set V. Now, consider the set $W = W_1 \cap W_2 \cap U \cap V$. Clearly, $W \subset U_1 \cap U_2$ and $W \subset V_1 \cap V_2$, and so

$$f_1|_W = f_2|_W$$
$$g_1|_W = g_2|_W$$

and from here, we see that for addition, we have

$$(f_1|_{W_1} + g_1|_{W_1})|_W = f_1|_W + g_1|_W = f_2|_W + g_2|_W = (f_2|_{W_2} + g_2|_{W_2})|_W$$

and for multiplication, we have

$$(f_1|_{W_1} \cdot g_1|_{W_1})|_W = f_1|_W \cdot g_1|_W = f_2|_W \cdot g_2|_W = (f_2|_{W_2} \cdot g_2|_{W_2})|_W$$

Therefore $(f_1|_{W_1} + g_1|_{W_1}, W_1)$ and $(f_2|_{W_2} + g_2|_{W_2}, W_2)$ are part of the same equivalence class $[f] + [g]$, and $(f_1|_{W_1} \cdot g_1|_{W_1}, W_1)$ and $(f_2|_{W_2} \cdot g_2|_{W_2}, W_2)$ are part of the same equivalence class $[f] \cdot [g]$. Addition and multiplication are commutative since

$$\begin{gathered}(f_1|_{W_1} + g_1|_{W_1})(x) = f_1|_{W_1}(x) + g_1|_{W_1}(x) \\ = g_1|_{W_1}(x) + f_1|_{W_1}(x) = (g_1|_{W_1} + f_1|_{W_1})(x)\end{gathered}$$

and

$$\begin{gathered}(f_1|_{W_1} \cdot g_1|_{W_1})(x) = f_1|_{W_1}(x) \cdot g_1|_{W_1}(x) \\ = g_1|_{W_1}(x) \cdot f_1|_{W_1}(x) = (g_1|_{W_1} \cdot f_1|_{W_1})(x)\end{gathered}$$

Finally, let $c \in \mathbb{R}$, $[f] \in C_p^\infty$, and $(f, U) \in [f]$. Then $c[f]$ is defined to be the equivalence class of all pairs equivalent to (cf, U). Scalar multiplication is well-defined because for pairs $(f_1, U_1), (f_2, U_2) \in [f]$, there exists a set $U \subset U_1 \cap U_2$ such that $f_1|_U = f_2|_U$ and so $cf_1|_U = cf_2|_U$. Therefore, (cf_1, U_1) and (cf_2, U_2) are part of the same equivalence class.